

EDUCATION

HEC Lausanne

M.Sc. Finance (Quantitative), GPA: 4.9/6.0

Lausanne, Switzerland
2019 – 2021

- **Thesis:** Rough Volatility Modeling in option pricing (6.0/6.0)

HEC Lausanne

B.Sc. Economics, GPA: 4.6/6.0

Lausanne, Switzerland
2016 – 2019

WORK EXPERIENCE

MyLift

Founding Engineer

Dublin, Ireland

February 2026 – Present

- Led end-to-end development of an Elixir/Phoenix medical tourism platform connecting French-speaking patients (France, Belgium, Switzerland) with 8 verified Tunisian doctors and 5 clinics across 10+ procedures and specialties. Owned architecture and product strategy.
- Integrated Stripe for split payments across multiple care providers and Daily.co for real-time video consultations, enabling cross-border transactions between European patients and care providers in Tunisia.
- Built role-based dashboards for doctors, clinics and patients along with an internal admin tool that powers the founding team's onboarding workflow for healthcare providers (clinics and doctors).
- Implemented direct client-to-S3 uploads via pre-signed URLs for galleries, profile images and leveraged a CDN to optimize media delivery performance.

Gartner

Software Engineer

Barcelona, Spain

January 2024 – November 2025

- Developed a GraphQL Elixir API serving over 100 million users annually, implementing a Query Specification pattern across PostgreSQL and Elasticsearch for composable query construction and consistent pagination.
- Rebuilt ad monetization interface, consolidating fragmented data sources and eliminating inconsistencies that disrupted ad delivery. This interface supports \$100M+ in annual revenue.
- Accelerated Kubernetes rollouts by 25% through implementing Amazon ECR layer caching, significantly reducing container deployment times for the engineering team.
- Built query complexity framework letting backend teams validate algorithm changes pre-rollout, preventing client breakage and doubling iteration speed on a previously rarely-touched algorithm.
- Built Kafka consumers across multiple Confluent topics handling 1M+ data points/day, processing both Avro and plain-format messages with schema-based deserialization where applicable.

Amazon

Software Developer

Iasi, Romania

May 2022 – October 2023

- Built a Python application to query product attributes (ASINs, etc.) on AWS Redshift, streamlining manual review for a 20 person team.
- Created a Python application enabling a team of associates to verify item inventory within fulfillment centers (10 people).
- Developed a Python application to automate the generation of batch vendor invoices from Excel spreadsheets, reducing redundant tasks for a team of associates working on billings (5 people).

PROJECTS

LiveDisplaced: Data visualization platform built upon Python and AWS services (ECS, Batch) to track global refugee movements from the United Nations data sources.https://github.com/EM51641/livedisplaced_

Async-sendgrid: Developed and maintained async-sendgrid, an open-source Python client for non-blocking email delivery on SendGrid, with 400,000+ PyPI downloads.<https://pypi.org/project/sendgrid-async/>

LANGUAGES

French, Arabic (native); English (C1); Spanish (B1)

SKILLS

Languages: Elixir, Python, TypeScript/JavaScript, Java, Nix

Backend & Data: Phoenix/LiveView, GraphQL, Postgres, Elasticsearch, Kafka, Redshift

Infra & Cloud: AWS (ECS/Fargate, EKS, Lambda, Aurora, S3, Batch), Terraform, Docker, Kubernetes

CI/CD & Observability: GitLab CI, GitHub Actions, SaltStack, Datadog, Prometheus, Grafana, OpenTelemetry